lecture7 Practice Questions and Solutions

Question 1 (q1_sol.py)

Pat, Chris, Blake, and Robin all love dolls. Pat and Chris have some number of dolls (entered by the user), while the number of dolls Blake and Robin have follow these rules:

- Blake has two dolls, plus three times as many as Pat.
- Robin has one more doll than whoever of Pat and Chris has the most dolls.

Write a program that gets the number of dolls for Pat and Chris, and then prints how many dolls each person has, plus the total number of dolls overall.

```
Dolls for Pat: 3
Dolls for Chris: 5

Pat has 3 dolls.
Chris has 5 dolls.
Blake has 11 dolls.
Robin has 6 dolls.
Total number of dolls: 25
```

Sample Solution

```
# get the number of dolls for Pat and Chris
pat_dolls = int(input("Dolls for Pat: "))
chris_dolls = int(input("Dolls for Chris: "))

# calculate the number of dolls for Blake and Robin
blake_dolls = 2 + 3 * pat_dolls
robin_dolls = 1 + max(pat_dolls, chris_dolls)
total_dolls = pat_dolls + chris_dolls + blake_dolls + robin_dolls

# print the number of dolls for each person
print()
print(f"Pat has {pat_dolls} dolls.")
print(f"Chris has {chris_dolls} dolls.")
print(f"Blake has {blake_dolls} dolls.")
print(f"Robin has {robin_dolls} dolls.")
print(f"Total number of dolls: {total_dolls}")
```

Question 2 (q2_sol.py)

Write a program that reads 3 ints from the user, and then prints then in order from smallest to biggest:

```
1st int: 5
2nd int: 0
3rd int: 2

In order: 0, 2, 5
```

Your program should work for any three integers, and it should only use features covered so far in the course, like basic arithmetic, printing, etc. Hint: Research Python's min and max functions.

Sample Solution

```
a = int(input('1st int: '))
b = int(input('2nd int: '))
c = int(input('3rd int: '))

first = min(a, b, c)
third = max(a, b, c)

# there are other ways to calculate second, but they are tricky
second = a + b + c - first - third

print(f'In order: {first}, {second}, {third}')
```

Question 3 (q3_sol.py)

Using turtle graphics, draw three equilateral triangles with the same center: a big one, a medium one inside the big one, and small one inside the medium. See threeTriangles.gif for example output.

Hint: turtle.up() lifts the turtle pen up so it doesn't draw a line when it moves. turtle.down() puts it back down.

Sample Solution

```
import turtle

# triangle 1
turtle.forward(300)
turtle.left(120)
turtle.forward(300)
turtle.left(120)
turtle.forward(300)
turtle.left(120)

# move turtle to the right and up
turtle.up()
turtle.forward(25)
turtle.left(90)
turtle.forward(25)
turtle.right(90)
turtle.forward(25)
turtle.down()

# triangle 2
turtle.forward(200)
turtle.left(120)
turtle.forward(200)
turtle.left(120)
turtle.forward(200)
turtle.left(120)

# move turtle to the right and up
turtle.up()
turtle.forward(25)
turtle.left(90)
turtle.forward(25)
turtle.right(90)
turtle.forward(25)
turtle.down()

# triangle 3
turtle.forward(100)
turtle.left(120)
turtle.forward(100)
turtle.left(120)
turtle.forward(100)
```

```
turtle.left(120)

turtle.done()
```